

NEXUS Engineering Physics: Module 2

Rotational Fields, Stokes' Theorem, and Magnetism

2.1 Circulation and the Curl of a Vector Field

The curl of a vector field measures its infinitesimal rotation or circulating tendency around a point. Given a vector field $\mathbf{B}$, the curl is denoted by $\text{curl } \mathbf{B}$ or $\nabla \times \mathbf{B}$.

2.2 Stokes' Theorem

Stokes' Theorem relates the line integral of a vector field around a closed boundary curve C to the surface integral of the curl over any surface S bounded by C .

Statement: If S is an oriented surface with piecewise smooth boundary curve C traversed in the positive direction, then $\oint_C \mathbf{A} \cdot d\mathbf{r} = \int_S (\text{curl } \mathbf{A}) \cdot d\mathbf{S}$.

Stokes' theorem generalizes the fundamental theorem of calculus to circulation and curl. It is the mathematical backbone of Ampere's law in integral form.

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2.3 Ampere's Circuital Law and Maxwell's Correction

Ampere's Circuital Law in integral form states that the line integral of magnetic field B around a closed loop equals μ_0 times the electric current passing through the loop:

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Applying Stokes' theorem converts the left-hand side into a surface integral of $\text{curl } B$. Maxwell identified an inconsistency for time-varying fields and added the displacement current term:

$$\text{curl } B = \mu_0 J + \mu_0 \epsilon_0 \frac{dE}{dt}$$

The second term is the displacement current density. This completed Maxwell's equations and predicted the propagation of electromagnetic waves in vacuum.